MIL-E-62170(AT)
10 May 1973

SUPERSEDING
ATPD-4032A
6 October 1969

MILITARY SPECIFICATION

ENGINES, GASOLINE: 4 AND 6-CYLINDERS HORIZONTAL OPPOSED AND 12 CYLINDER, 90° V-TYPE (OVERHAUL)

1. SCOPE

1.1 Scope. This specification covers requirements for 4 and 6-cylinder, horizontal opposed, air-cooled, 4-stroke-cycle, fuel injection or carbureted, supercharged or non-supercharged, and 12-cylinder, 90-degrees V-type, air-cooled 4-stroke-cycle, fuel injection or carbureted, supercharged or non-supercharged internal-combustion, overhauled gasoline engines (see 6.1).

1.2 Classification. Engines shall be type classified as follows (see 6.2):

Engine, Gasoline:	AOI-268-3B, 127 hp.
Engine, Gasoline:	AOI-402, 200 hp.
Engine, Gasoline:	AO-895-4, 375 hp.
Engine, Gasoline:	AOS-895-3, 500 hp.
Engine, Gasoline:	AOSI-895-5, 500 hp.
Engine, Gasoline:	AV-1790-5 810 hp.
Engine, Gasoline:	AV-1790-7, 810 hp.
Engine, Gasoline:	AV-1790-7B, 810 hp.
Engine, Gasoline·	AV-1790-7C 810 hp.
Engine, Gasoline:	AVI-1790-8, 810 hp.
Engine, Gasoline:	AVSI-1790-6, 1020 hp.
Engine, Gasoline:	AVSI-1790-6A, 1000 hp.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of the specification to the extent specified herein.

FSC-2805

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS
Military

MIL-P-514 - Plates, Identification, Instruction and Marking Blank.
MIL-L-2104 - Lubricating Oil, Internal-Combustion Engine, Tactical Service
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-A-13488 - Air Cleaner, Engine· Heavy Duty, Oil-Bath Type (for Internal-Combustion Engines).
MIL-E-13856 - Electrical Components for Automotive Vehicles· Waterproofness Tests.
MIL-F-13927 - Fungus Resistance Test, Automotive Components.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative and Break-in.
MIL-S-45005 - Seal, Plain and Seal, Plain, Encased: Fluid, Radial, Single and Multiple Lip Sealing Member.
MIL-G-46015 - Gasoline, Automotive, Combat: Referee Grade.

STANDARDS
Military

MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

PUBLICATIONS

The List of Drawings (LD), List of Parts Lists (LP), List of Specifications (LS), Master List of Supplementary Quality Assurance Provisions (QM), and associated lists appearing on the Engineering Index List (IL) shall be used as applicable for each engine in Table I.

Table I. Engineering drawings and associated Lists

IL-7388574 - Engine, Gasoline: AO-895-4, 375 hp.
IL-7355952 - Engine, Gasoline: AV-1790-7, 810 hp.
IL-7527948 - Engine, Gasoline: AOS-895-3, 500 hp.
IL-7721825 - Engine, Gasoline: AV-1790-5, 810 hp.
IL-8365416 - Engine, Gasoline: AVSI-1790-6, 1,020 hp.
IL-8376655 - Engine, Gasoline: AOSI-895-5, 500 hp.
IL-8376827 - Engine, Gasoline: AV-1790-7B, 810 hp.
IL-8376947 - Engine, Gasoline: AV-1790-7C, 810 hp.
IL-8380568 - Engine, Gasoline: AVI-1790-8, 810 hp.
IL-8732609 - Engine, Gasoline: AOI-268-3B, 127 hp.
IL-8745822 - Engine, Gasoline: AOI-402, 200 hp.
IL-8745836 - Engine, Gasoline: AVSI-1790-6A, 1000 hp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by overhaul contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First overhauled engine. A complete overhauled engine to be furnished under this specification shall be produced prior to the overhauling of engines in quantity. The complete overhauled engine shall be submitted to the Government for examinations and tests to determine conformance to this specification. The first overhauled engine submitted by the contractor shall be fully representative of engines proposed to be supplied by the overhaul contractor from his manufacturing facilities and tooling.

3.1.2 Interchange parts. New parts and subassemblies required for the overhaul of engines specified in Table I shall be constructed to permit their interchange and installation without modification. A matched set of parts shall be interchangeable as a complete subassembly. Parts for a selected fit shall be as specified.

3.2 Reliability. After proper installation into the vehicle for which the engine is specified, the engine shall require no repair and no maintenance other than replacing spark plugs, ignition points, oil filter, and air cleaner during engine operations up to 200 hours.

3.3 Materials. All materials, parts, and assemblies used in the overhaul of engines shall conform to requirements of applicable drawings, specifications, and standards and shall be free from defects that adversely affect the serviceability and function of the engine.

3.3.1 Qualified products list (QPL). The contractor shall be responsible for using all assemblies and parts necessary in engine overhaul that are listed as qualified products on the QPL and those approved for inclusion on the QPL. Assemblies and parts determined as QPL items appear on the applicable Engineering List of Specifications cited in Table I. The contractor's manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products but not yet listed on the QPL, the contractor shall list the products by number and date of the approval document with the name of supplier(s).

3.3.2 Electrical components. All electrical components, including wiring and electrical connections, shall conform to the applicable requirements of MIL-E-13856 and MIL-F-13927. Unless otherwise specified, all electrical components shall be installed on the engine and properly adjusted.

3.3.3 Oil seals. All new oil seals used in overhauled engines shall conform to MIL-S-45005.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.3.4 Air cleaners. All air cleaners for overhauled engines shall conform to MIL-A-13485.

3.4 Engine overhaul. Engine shall be overhauled in accordance with this specification. Parts and assemblies shall conform to applicable documents specified herein. All parts requiring identification shall be in accordance with MIL-STD-130.

3.4.1 Overhauled engine endurance run. The first engine of each type overhauled shall be submitted to an endurance run to determine conformance to operational requirements of this specification. The bare engine (see 6.4), using fuel conforming to grade 1 of MIL-G-46015 and grade of oil conforming to seasonal requirements of MIL-L-2104, shall be operated for a period of 200 hours as specified in Table II. At conclusion of the 200 hours run the engine shall have met specified performance requirements and shall develop not less than 90 percent of its rated horsepower as shown in Table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table II. Endurance run

Period	Time in hours	RPM or percent of max. rated speed	Percent of full-throttle load at indicated speed	Cylinder head temp. °F.	Cylinder base temp. °F.
1	2	(Break-in run, see 3.5)			
2	2	650 rpm	0	See 3.4.2.6	See 3.4.2.6
3	5	50 pct	75	do	do
4	1	650rpm	0	do	do
5	5	75 pct	75	do	do
6	1	650 rpm	0	do	do
7	5	85 pct	75	do	do
8	1	650 rpm	0	do	do
9	5	100 pct	75	do	do
10	2	650 rpm	0	do	do
11	5	50 pct	75	do	do
12	1	650 rpm	0	do	do
13	5	75 pct	75	do	do
14	1	650 rpm	0	do	do
15	5	85 pct	75	do	do
16	1	650 rpm	0	do	do
17	5	100 pct	75	do	do
18	2	50 pct	50	do	do
19	1	650 rpm	0	do	do
20	2	50 pct	100	do	do
21	2	50 pct	50	do	do
22	1	650 rpm	0	do	do
23	2	50 pct	100	do	do
24	4	75 pct	100	do	do
25	1	650 rpm	0	do	do
26	4	75 pct	50	do	do
27	1	650 rpm	0	do	do
28	4	75 pct	100	do	do
29	1	650 rpm	0	do	do
30	4	75 pct	50	do	do
31	4	85 pct	75	do	do
32	1	650 rpm	0	do	do
33	4	85 pct	100	do	do
34	1	650 rpm	0	do	do
35	4	85 pct	75	do	do
36	1	650 rpm	0	do	do
37	4	100 pct	100	do	do
38	1	650 rpm	0	do	do
39	4	100 pct	75	do	do

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table II. Endurance run (Cont'd)

Period	Time in hours	RPM or percent of max. rated speed	Percent of full-throttle load at indicated speed	Cylinder head temp. °F.	Cylinder base temp. °F.
40	See 3.4.2.4 and 3.4.2.5 (Power and torque tests)				
41	1	650 rpm	0	do	do
42	2	105 pct	100	do	do
43	1	650 rpm	0	do	do
44	2	105 pct	100	do	do
45	2	105 pct	75	do	do
46	1	650 rpm	0	do	do
47	2	105 pct	100	do	do
48	2	105 pct	75	do	do
49	1	650 rpm	0	do	do
50	2	105 pct	100	do	do
51	2	105 pct	75	do	do
52	1	650 rpm	0	do	do
53	2	105 pct	100	do	do
54	2	75 pct	75	do	do
55	1	650 rpm	0	do	do
56	2	105 pct	100	do	do
57	2	75 pct	75	do	do
58	1	650 rpm	0	do	do
59	2	85 pct	100	do	do
60	2	100 pct	100	do	do
61	1	650 rpm	0	do	do
62	2	75 pct	75	do	do
63	2.5	85 pct	100	do	do
64	2.5	85 pct	75	do	do
65	2.5	85 pct	100	do	do
66	2.5	85 pct	75	do	do
67	2.5	100 pct	100	do	do
68	2.5	100 pct	75	do	do
69	2.5	100 pct	100	do	do
70	2.5	100 pct	75	do	do
71	0.75	105 pct	100	do	do
72	0.25	650 rpm	0	do	do
73	0.75	100 pct	100	do	do
74	0.25	650 rpm	0	do	do
75	0.75	105 pct	100	do	do
76	0.25	650 rpm	0	do	do
77	0.75	105 pct	75	do	do
78	0.25	650 rpm	0	do	do
79	0.75	105 pct	100	do	do
80	0.25	650 rpm	0	do	do

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table II. Endurance run (Cont'd)

Period	Time in hours	RPM or percent of max. rated speed	Percent of full-throttle load at indicated speed	Cylinder head temp. °F.	Cylinder base temp. °F.
81	0.75	105 pct	100	do	do
82	0.25	650 rpm	0	do	do
83	0.75	105 pct	100	do	do
84	0.25	650 rpm	0	do	do
85	0.75	105 pct	100	do	do
86	0.25	650 rpm	0	do	do
87	0.75	105 Pct	75	do	do
88	0.25	650 rpm	0	do	do
89	0.75	105 pct	100	do	do
90	0.25	650 rpm	0	do	do
91	0.75	105 pct	100	do	do
92	0.25	650 rpm	0	do	do
93	0.75	105 pct	100	do	do
94	0.25	650 rpm	0	do	do
95	0.75	105 pct	100	do	do
96	0.5	650 rpm	0	do	do
97	0.75	105 pct	100	do	do
98	0.5	650 rpm	0	do	do
99	0.75	105 pct	100	do	do
100	0.5	650 rpm	0	do	do
101	0.75	105 pct	100	do	do
102	0.5	650 rpm	0	do	do
103	0.75	105 pct	100	do	do
104	0.5	650 rpm	0	do	do
105	0.75	105 pct	100	do	do
106	0.5	650 rpm	0	do	do
107	0.75	105 pct	100	do	do
108	0.5	650 rpm	0	do	do
109	0.75	105 pct	100	do	do
110	0.5	650 rpm	0	do	do
111	0.75	105 pct	100	do	do
112	0.5	650 rpm	0	do	do
113	0.75	105 pct	100	do	do
114	0.5	650 rpm	0	do	do
115	0.75	105 pct	100	do	do
116	0.5	650 rpm	0	do	do
117	0.75	105 pct	100	do	do
118	0.5	650 rpm	0	do	do
119	0.75	105 pct	100	do	do
120	0.5	650 rpm	0	do	do
121	10.	105 pct	100	do	do
122	See 3.4.2.4 and 3.4.2.5 (Power and torque tests)				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.4.1.1 Recorded data. The contractor shall furnish certified graphs with performance curves showing horsepower, torque, and fuel and oil consumption of each overhauled engine submitted to the endurance run. The contractor shall maintain a log sheet (see 6.2) for each engine and record the data from each run in accordance with Table II.

3.4.1.2 Physical evaluation. At completion of the endurance run, all critical parts of the overhauled engine shall be evaluated to determine the degree of changes in dimensions and physical or chemical characteristics.

3.4.2 Performance requirements. The overhauled engine shall meet all performance requirements under all loads and conditions specified herein.

3.4.2.1 Performance runs. Performance runs of the engine shall be made when stabilized operating conditions are obtained as follows:

(a) Engine shall run until torque, speed, and temperature have been maintained without noticeable change for a period of not less than 1 minute.
(b) Each run shall be for a period of 3 minutes or longer.

3.4.2.2 Governor. The governor shall be adjusted to limit the engine speed as specified in Table III. The engine speed shall stabilize within 10 seconds after throttle position is reached. The governor shall be sealed after adjustment.

Table III. Governor adjustment

Rated hp	Full load operation, rpm	No load operation maximum rpm
127	3200 to 3250	3350
200	3000 to 3100	3250
375	2800 to 2850	3000
500	2800 to 2850	3000
810	2800 to 2850	3000
1,000	2800 to 2850	3000
1,020	2800 to 2850	3000

3.4.2.3 Speed range. The engine shall meet performance requirements under all loads and conditions as specified herein through the speed ranges specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table IV. Speed ranges

BHP	Speed range rpm	Maintained idle - rpm
127 ----------	1000 to 3200	650
200 ----------	1200 to 3000	650
375 ----------	1200 to 2800	650
500 ----------	1200 to 2800	650
810 ----------	1200 to 2800	650
1,000 ----------	1200 to 2800	650
1,020 ----------	1200 to 2800	650

3.4.2.4 Gross horsepower. The bare engines (see 6.4), shall develop the gross horsepower specified in Table V

Table V. Gross horsepower

Rated hp	AT rpm
127 minus 5% ----------	3200
200 minus 5% ----------	3000
375 minus 2% ----------	2800
500 minus 2% ----------	2800
810 minus 2% ----------	2800
1,000 minus 2% ----------	2800
1,020 minus 2% ----------	2800

3.4.2.5 Gross torque. The bare engines (see 6.4), under full throttle shall develop the gross torque at speeds specified in Table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table VI. Gross torque

Rated hp	Torque lbs-ft	At rpm
127	214 minus 5%	2900
200	347 minus 5%	2800
375	775 minus 2%	2000
500	945 minus 2%	2400
810	1,000 minus 2%	2000
1,000	1,820 minus 2%	2000
1,020	1,875 minus 2%	1900

3.4.2.6 Limiting operating temperature. The engine shall not exceed the following limiting temperatures when operating at full load throughout its speed range under any of the conditions specified herein:

(a) Oil temperature. Oil temperature, measured at the oil-cooler outlet, when operating with oil coolers and with a maximum air-inlet temperature of 115°F. shall be no more than 240°F.

(b) Cylinder-head temperatures. Cylinder-head temperature, measured on a spark plug gasket, shall be not more than 525°F. under any condition operation specified herein.

(c) Cylinder-base temperature. Cylinder-base temperature, measured on the downstream centerline, 1 inch from the top of the mounting flange, shall be not more than 350°F. under any condition of operation specified herein.

3.4.2.7 Fuel consumption. The bare engine (see 6.4) using fuel as specified herein and operating at full throttle and under full load shall consume fuel per gross brake-horsepower of not more than the amounts specified in Table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table VII. Fuel consumption

Rated hp	Lb/bhp/hr max.	At rpm
127 ----------	0.62	3200
200 ----------	0.61	3000
375 (Carbureted) -----	0.62	2800
500 (Fuel injector) --	0.645	2800
500 (Carbureted) -----	0.73	2800
810 (Fuel injector)---	0.55	2800
810 (Carbureted)------	0.60	2800
1,000 (Fuel injector) --	0.67	2800
1,020 (Fuel injector) --	0.67	2800

3.4.2.8 Oil consumption. The engines shall consume not more than 0.015 pounds of lubricating oil per gross brake-horsepower per hour, when operating at full throttle throughout the speed range of the engine, using oil as specified herein.

3.4.2.9 Oil pressure. Using oil as specified herein, with a temperature of plus 140 to plus 150°F. gallery oil pressure, measured at the pressure sending unit, pressures shall be as specified in Table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table VIII. Oil pressures

Model	BHP	Not more than-psi	Nor less than-psi	Specs	Grade of oil used	Engine speed-rpm
AOI-268-3B	127	75	50		50	3200
			15	MIL-L-2104	50	650
		75	30	or	30	3000
			15	MIL-L-21260	30	650
AOI-402	200	75	60		50	3000
			15	MIL-L-2104	50	650
		75	30	or	30	3000
			15	MIL-L-21260	30	650
AOS-895-3	375	80	65		50	2800
			15	MIL-L-2104	50	650
		80	45	or	30	2800
			15	MIL-L-21260	30	650
AO-895-4	375	80	65		50	2800
			15	MIL-L-2104	50	650
		80	45	or	30	2800
			15	MIL-L-21260	30	650
AOSI-895-5	500	80	65		50	2800
			15	MIL-L-2104	50	650
		80	45	or	30	2800
			15	MIL-L-21260	30	650
AV-1790-5	810	80	65		50	2800
			15		50	650
AVSI-1790-6	1020	80	45		30	2800
AVSI-1790-6A	1000	80	45	MIL-L-2104	30	2800
			15	or	30	650
AV-1790-7	810	80	65	MIL-L-21260	50	2800
AV-1790-7B	810	80	15		50	650
AV-1790-7C	810	80	45		30	2800
AVI-1790-8	810	80	15		30	650

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.4.2.10 Submersion. The engine, with intake and exhaust ducted to the atmosphere, shall be capable of operating for a period of 30 minutes while submerged to a depth of 60 inches above the cooling fans in either fresh or salt water. After said operation and while submerged, the engine shall restart after being stopped for three minutes, and shall be operated for an additional 15 minutes.

3.4.2.10.1 Water contamination. A maximum of 2 percent water contamination in the lubricating oil is permissible after being subjected to submersion conditions.

3.4.2.10.2 Air pressure drop. With internal air pressure of 5 psi applied, air pressure drop shall not be more than 1 3/4 psi in a 3 minute period.

3.4.2.11 Electromagnetic compatibility. The overhauled engine shall meet the electromagnetic control requirements for class 11C and class III equipment specified in MIL-STD-461.

3.5 Overhauled engine break-in run. Each overhauled engine shall be subjected to a break-in run by the contractor before acceptance by the Government. Each engine shall be coupled with a dynamometer or other variable-loading device and shall be operated in accordance with Table IX. During the engine break-in runs, the engine shall perform as specified herein using fuel in accordance with seasonal requirements of MIL-G-3056. Lubricating oil shall be used in accordance with the seasonal requirements of MIL-L-2104. The governor shall be adjusted and sealed as specified in 3.4.2.2 and Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table IX. Break-in run

Engines by rated hp	Run	Time, minutes	RPM		Absolute manifold pressure, in. Hg.	
			127	200	127	200
127 and 200	1	5	1200	1200	14.0	15.5
	2	5	1600	1600	17.5	16.5
	3	5	2000	2000	18.2	19.0
	4	5	2400	2400	20.3	23.0
	5	15	2800	2800	Full throttle and full load	
	6	15	3200	3000		
	7	5	2600	2600	do	
	8	5	2000	2200	do	
	9	5	1400	1800	do	
	10	5	Idle speed			

375,500, 810 1000 and 1020			375, 500, 810 1000 and 1020	Percent of bhp
	1	15	1000	5
	2	20	1400	20
	3	20	1800	30
	4	20	2200	60
	5	30	2400	70
	6	30	2600	80
	7	30	2800	Full throttle
	8	5	2400	Full throttle
	9	5	2000	Full throttle
	10	5	1600	Full throttle
	11	5	Idle speed	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.5.1 **Break-in run recorded data.** The contractor shall maintain a log sheet (see 6.2) if each engine during the break-in run in accordance with Table IX.

3.6 Exterior surface treatment. All exposed exterior surfaces of the engine and its components shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with applicable provisions of MIL-STD-193.

3.7 Identification plates. Unless otherwise shown on the applicable drawings, the nameplate shall conform to MIL-P-514.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's inspection or quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current, written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall notify the Government of any change to the written procedure subsequent to submission of procedure to the contracting officer.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiences discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.1.3 First overhauled engine. To conform to 3.1, the first overhauled engine of each type shall be submitted to the Government for examinations and tests to assure conformance to this specification and to establish a standard for all subsequent engines to be overhauled under provisions of this specification and procurement contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.1.4 Interchangeability of parts check. To conform to 3.1.2 parts and subassemblies shall be as specified in applicable drawings and specifications. Parts and subassemblies shall be inspected to applicable drawings and documents listed in Table I.

4.2 Reliability. To determine conformance to 3.2, the first overhauled engine of each type shall be subjected to tests as required by the specific engine reliability requirements.

4.3 Materials. To conform to 3.3, all materials, parts, and assemblies shall be as specified.

4.3.1 Qualified products. The overhaul contractor's records shall be checked to determine conformance to 3.3.1 and that all QPL items shown on the applicable List of Specifications data sheets are used.

4.3.2 Accessories and equipment check. To conform to 3.3.2 all electrical accessories and equipment shall be as specified.

4 4.3.3 Oil seals. In accordance with 3.3.3, all new oil seals used in overhauled engines shall conform to MIL-S-45005.

4.3.4 Air cleaners. To conform to 3.3.4, all air cleaners used on overhauled engines shall conform to MIL-A-13488.

4.4 Engine overhaul. To conform with 3.4, all overhauled engines shall satisfy all requirements as specified herein. The contractor shall maintain and provide specified data of engine runs and tests to the Government.

4.4.1 Overhauled engine endurance run test. To determine conformance to 3.4 through 3.4.1.2, the first engine of each type overhauled shall be operated as specified in Table II, using fuel and oil as specified herein. The engine shall be subjected to examinations and tests specified in Table X and XI. The endurance run, examinations and tests shall be performed at a place designated by the Government. The endurance run, examinations and tests shall be performed by the Government. Delay in completion of the test caused by engine breakdown or failure of the contractor to support the engine with parts during the test shall not be basis for adjustment of the contract, delivery schedule or contract price. The endurance run shall be continuous and running time less than 30 minutes shall not be counted toward fulfillment of endurance hours. Test data required shall be as specified in 3.4.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table X. Check list for visual examinations

Item	Visual defects
Engine	Malfunction, improper adjustment or installation of components
Valve tappets	Sticking valves. Improper clearance (see Note end of Table)
Air cleaner	Omitted, loose, or damaged
Oil and fuel filter	Omitted, improperly installed, or damaged
Spark plugs	Broken, loose, improper setting
Ignition points	Improper setting
Governor	Malfunction, improper linkage
Cylinder Base, stud, manifold bolts, and crank-case bolts	Improper torque
Fuel and oil lines	Damaged, leakage, plugged.
Hose	Damaged, improperly installed
Drain plugs	Omitted or loose
Data plates	Omitted or incorrect marking
Belts	Loose, damaged
Minor assemblies	Omitted, damaged, improperly installed or adjusted
Wiring	Omitted, loose, damaged, or improperly installed

NOTE: Valve tappet clearance shall be as follows:

(a) Intake(cold) 0.010 inch
(b) Exhaust (cold) 0.025 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table IX. Test and examination reference paragraphs

Item	First Engine	Break-in Engines	Acceptance Engines	Controls Engines
Engine runs	4.4.1	4.5.1	4.5.3	4.6.1
Tear-down	4.4.1.1	4.5.2	-	4.6.7
Examinations	4.4.1.1	4.5.2	4.5.3	4.6.4
Gaging	4.4.1.2	-	-	4.6.7
Performance	4.4.2	4.5	4.5.3	4.6.5
Power	4.4.2.2	4.5	4.5.3	4.6.5
Fuel consumption	4.4.2.3	4.5	4.5.3	4.6.5
Submersion	4.4.2.4	4.5	4.5.3	4.6.5
Oil contamination	4.4.2.4.1	4.5	4.5.3	4.6.5
Air pressure	4.4.2.4.2	4.5	4.5.3	4.6.5
Electromagnetic	4.4.2.5	4.5	4.5.3	4.6.5
Interchangeability	4.1.4	4.1.4	4.1.4	4.1.4
Reliability	4.2	4.2	4.2	4.2
Materials	4.3	4.3	4.3	4.3
QPL	4.3.1	4.3.1	4.3.1	4.3.1
Accessories	4.3.2	4.3.2	4.3.2	4.3.2
Oil seals	4.3.3	4.3.3	4.3.3	4.3.3
Air cleaners	4.3.4	4.3.4	4.3.4	4.3.4
Recorded data	4.7.1	4.7.1	4.7.1	4.7.1
Test equipment	4.8	4.8	4.8	4.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.4.1.1 Endurance run teardown inspection. To determine conformance to 3.4.1 through 3.4.1.2, at the conclusion of the endurance run the contractor, in presence of the Government inspector, shall disassemble the engine. The disassembly shall be accomplished to the extent required to perform the following inspections:

a. Contamination of oil
b. Dirt and metal chips in engine assembly, oil pan and filter
c. Torque of main-bearing bolt
d. Torque of connecting-rod bolts
e. Scuffing, scoring, galling of cylinder
f. Scuffing, scoring, galling of supercharger drive
g. Scuffing, scoring, galling of shaft, gears, and clutch of cooling fan assembly

4.4.1.2 Gaging and magnetic-powder testing. To determine conformance to 3.4.1.2, the critical parts of the disassembled engine shall be inspected by gaging or magnetic-power testing. The results of the inspection shall be recorded to enable comparison between the physical dimensions and characteristics of parts of the overhauled and those of the same type engine originally procured as a new engine. The following parts, or areas of parts, shall be inspected as specified.

(a) Cylinder bores. Both parallel and perpendicular to the crankshaft, at the top, center, and bottom of track of ring travel above the piston pins.

(b) Pistons. Diameters above the topmost ring, parallel and perpendicular to the piston pin, and at center of skirt, perpendicular to the piston pin.

(c) Piston rings.
1. Gap clearance in a standard ring gage, or when fitted in an unused cylinder.
2. Free gap.
3. Clearance of ring in groove.
4. Depth of ring groove.
5. Width of ring groove.

(d) Crankshaft.
1. Main bearing journals.
2. Connecting rod journals
3. Alignment.
4. Magnetic-powder inspection.

(e) Main-bearing inserts.
1. Assembled diameters.

(f) Connecting-rod bearings.
1. Assembled diameters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

(g) Piston pins.
1. Diameters and fits

(h) Gears and gear trains.
1. Backlash.

(i) Valves, intake and exhaust.
1. Diameter of stem.
2. Out-of-roundness-head and stem.
3. Warpage.
4. Magnetic-powder inspection.

(j) Valve springs.
1. Free length.
2. Compressed tension(lb).
3. Magnetic-powder inspection.

(k) Valve guides, intake and exhaust.
1. Inside diameter.
2. Out-of-roundness.
3. Warpage.

(l) Valve-spring retainers.
1. Magnetic-powder inspection.

(m) Splines and spline drives.
1. Roll size.
2. Major diameter
3. Comp gage.
4. Magnetic-powder inspection.

4.4.1.3 Endurance run test failure. Any failure or deficiency of workmanship or materials found during or as a result of the endurance run will be cause for rejection of the engine. Further, the Government will cease acceptance of overhauled engines until objective evidence is provided by the contractor that corrective action has been taken to eliminate cause of the failure of deficiency. Any failures or deficiency found during or as a result of the endurance run shall be prima facie evidence that all engines accepted prior to completion of the endurance run are similarly deficient unless objective evidence is furnished by the contractor that they are not similarly deficient. Such deficiencies on all overhauled engines shall be corrected by the contractor at no cost to the Government.

4.4.2 Performance requirements check. To determine conformance with 3.4.2 through 3.4.2.11, the overhauled engine shall be operated and tested as specified herein and performance data shall be recorded to assure compliance with all specified requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.4.2.1 Performance runs check. To conform with 3.4.2.1, the engine shall be operated as specified and performance data shall be obtained under stabilized operating conditions. Because some engines require more running time than others to reach a stabilized condition, the following shall apply.

(a) No data shall be taken until torque, speed, and temperature have been maintained without noticeable change for a period of not less than 1 minute.
(b) Recorded data shall be the average sustained values maintained for a period of not less than 1/2 minute, with no significant change during that time.
(c) Each run shall be for a period of 3 minutes or longer.

4.4.2.2 Power test. To determine conformance to 3.4.2 and 3.4.2.5, engines shall be operated as specified. Prior to the power test, the lubricating oil shall be changed and all filters serviced, and spark plugs, ignition points, timing and valve clearances shall be checked and adjusted to applicable requirements. Full throttle power and torque shall be determined with standard carburetor and spark or injection setting as applicable, at a minimum of eight speeds. Runs shall be approximately equally spaced over a range starting at the lowest steady speed at which the engine will operate, to 105 percent of the maximum rated speed only for 1790 cubic inch engines, and 110 percent only for engines under the 1790 cubic inches. All points from which performance curves are plotted shall be clearly shown on the certified graphs (see 3.4.1.1).

4.4.2.3 Fuel consumption check. To determine conformance to 3.4.2.7, the engine shall be operated as specified. Fuel consumption shall be measured simultaneously with brake horsepower at full throttle. The fuel consumption measurement shall not be started until the engine temperature, load, and rpm are stabilized for 1 minute. A measuring interval of not less than 1 minute but more than 40 seconds shall be used. All specific fuel consumption measurements shall be based on gross brake horsepower.

4.4.2.4 Submersion test. To determine conformance to 3.4.2.10, the engine shall be operated as specified and checked for compliance with specified requirements.

4.4.2.4.1 Oil contamination check. The lubricating oil in the engine shall be checked to assure water contamination is 2 % or less after submersion test.

4.4.2.4.2 Air pressure leakage test. An internal air pressure of 5 psi shall be supplied to the engine. When air is stabilized, the air valve shall be closed. The air pressure drop shall be observed to determine conformance to 3.4.2.10.2.

4.4.2.5 Electromagnetic compatibility check. To determine conformance to n3.4.2.11, the engine shall be checked in accordance with MIL-STD-461.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.5 · Overhauled engine break-in test. To determine conformance to 3.5, each overhauled engine shall be operated as specified and tested to assure meeting of specified requirements. During the break-in run there shall be no evidence of malfunction. The governor shall be set and tested in conformance to 3.4.2.2 and Table III.

4.5.1 Rerun break-in test. During the break-in test, any malfunction in e engine performance shall be corrected by the contractor. The engine shall be reassembled, if applicable, and shall be resubmitted to a break-in run in accordance with Table XII. Engine shall meet all specified break-in run requirements before acceptance by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table XII. Rerun break-in run

Engines by rated hp	Run	Time, minutes	RPM		Absolute manifold pressure, in. Hg.	
			127	200	127	200
127 and 200	1	5	1200	1200	14.0	15.5
	2	5	1600	1600	17.5	16.5
	3 3	5	200	2000	18.2	19.0
	4	5	400	2400	20.3	23.0
	5	15	2800	2800	Full throttle and full load	
	6	15	3200	3000		
	7	5	2600	2600	do	
	8	5	2000	2200	do	
	9	5	1400	1800	do	
	10	5	Idle speed -		-	

			375, 500, 810, 1020	Percent of bhp
375, 500, 810, 1000 and 1,020	1	10	1000	5
	2	15	1400	20
	3	15	1800	30
	4	15	2200	60
	5	15	2600	80
	6	15	2800	Full throttle and full load
	7	5	2400	
	8	5	2000	do
	9	5	1600	do
	10	5	Idle speed	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.5.2 Engine break-in tear-down inspection. At conclusion of the applicable break-in runs, the contractor, in presence of the Government inspector, shall disassemble and inspect the first 5 overhauled engines. Engine disassembly shall be accomplished to the extent required to perform the following inspections:

(a) Contamination of oil.
(b) Dirt and metal chips in engine assembly, oil pan and filter.
(c) Torque of main-bearing bolt.
(d) Torque of connecting-rod bolts.
(e) Scuffing, scoring, galling of cylinder.
(f) Scuffing, scoring, galling of supercharger drive.
(g) Scuffing, scoring, galling of shaft, gears, and clutch of cooling fan assembly.

4.5.2.1 Corrections and reassembly test. Should there be evidence of any condition of engine or parts requiring correction, such corrections shall be made by the contractor prior to reassembly of the engine. In addition, the contractor shall submit objective evidence to the Government, that corrective action has been taken in production to eliminate any deficiency disclosed. Failure of the contractor to submit such evidence shall be cause for refusal of the Government to continue acceptance of subsequent overhauled engines.

4.5.3 Acceptance tests. Before overhauled engines are accepted by the Government, each engine shall be subjected to examinations listed in Table X and shall be subjected to applicable tests specified in Table XI.

4.5.3.1 Idle speed check. After completion of acceptance tests, the linkage shall be adjusted to idle speed in conformance to Table IV.

4.6 Fifty-hour run control test. During the engine-overhaul contract period, a 50-hour run control test shall be conducted as specified herein for the purpose of determining the continued quality of overhauled engines.

4.6.1 Fifty-hour test run. The bare engine (see 6.2) shall be run in accordance with Table XIII. Fuel conforming to grade 1 of MIL-G-46015 and lubricating oil conforming to MIL-L-2104 shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

Table XIII. Fifty-hour control test run

Engines by rated hp	Time	Engine speed rpm at full load	
		127	200
127 and 200	20 Hours	3200	3000
	10 Hours	3520	3150
	20 Hours	3200	3000
	10 Minutes	2400	2600
	10 Minutes	2000	2200
	10 Minutes	1600	1800
375 and 500	20 Hours		2800
	10 Hours		3080
	20 Hours		2800
	10 Minutes		2400
	10 Minutes		2000
	10 Minutes		1600
810, 1000, and 1,020	20 Hours		2800
	10 Hours		2940
	20 Hours		2800
	10 Minutes		2400
	10 Minutes		2000
	10 Minutes		1600

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.6.2 Selection of test sample. Overhauled engines shall be elected at the rate of one per month when engines overhauled is 100 engines per month or less, and at the rate of two per month when engines overhauled is greater than 100 engines per month.

4.6.3 Pre-test warmup. All data shall be obtained under stabilized operating conditions, with engine coupled to a dynamometer. At each start, the engine shall be warmed up until torque, speed and temperature have been stabilized for a period of not less than 3 minutes. All applicable temperatures shall be recorded at the conclusion of each warmup time and running time of less than 30 minutes shall not be counted toward fulfillment of test run hours. Prior to test, all applicable settings and adjustments shall be reset to specifications before test is begun. Data shall be recorded at 1 hours intervals during test.

4.6.4 Test. The engine shall be examined for deficiencies specified in Table X and subjected to the tests specified for control in Table XI.

4.6.5 Engine performance. During the 50-hour test run the engine shall be observed for conformance to performance requirements specified in 3.4.2 through 3.4.2.11.

4.6.6 Control test failure. Failure of the engine to meet any of the tests or any evidence of abnormal wear such as scuffing or galling of the cylinder bores, pistons, piston rings, piston pins, bearings, bearings journals, gear teeth, cam surfaces, valves, tappets, or other parts, burning of valves or pistons, sticking of valves or piston rings, leakage of gaskets or seals, imminent failure of a component or other malfunctions, shall constitute a failure of the test. Failure of a control test sample to pass any specified examination or test shall be cause for the Government to refuse acceptance of overhauled engines until it has been proven to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.6.7 Corrections. Upon completion of the test the engine shall be completely disassembled and all parts subjected to wear or stress shall be inspected. A complete report including the inspection and testing of the engine shall be submitted to the Government inspector and a copy to the contracting officer. Should there be evidence of failure of parts as a result of this test, the inspector shall refuse acceptance of subsequent production engine until such corrections have been made and objective evidence presented to the Government inspector. The engine subjected to this test shall be reassembled, with new parts replacing those found unsatisfactory during examination.

4.6.8 Disposition of engine. After correction of all defects disclosed by the test, engine shall be reassembled, tested as specified for acceptance in Table XI, inspected, and if found acceptable, engine will be drained of all break-in oil and engine shall be preserved as specified and then be shipped.

4.7 Performance data. The contractor shall maintain a log sheet (see 6.2) for each engine during the endurance runs, break-in runs, and 50-hour control runs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.7.1 Recorded date. The following data shall be recorded at end of each hour of operation, at the completion of each period, and just prior to stopping the engine (emergency stops excluded).

(a) Engine speed, rpm.
(b) Engine load, bhp.
(c) Intake manifold pressure, in. Hg and exhaust manifold pressure, in. water.
(d) Lubricating oil pressure, (main), psi.
(e) Carburetor or intake air entrance pressure, in Hg.
(f) Crankcase pressure, psi.
(g) Lubricating oil temperature, °F. at oil cooler inlet.
(h) Lubricating oil temperature, °F. at oil cooler outlet.
(i) Lubricating oil flow if possible, gpm.
(j) Blow-by, cfm for:

1. Each hour on the 200 hour test.
2. At first and last hour of 50-hour control test.

(k) Fuel flow, lb/hr.
(l) Fuel pressure at carburator or injection pump, °F.
(m) Fuel temperature at carburator or injection pump, °F.
(n) Air temperature entering through oil coolers and air entrance grilles, °F.
(o) Air temperature between fan and engine, °F. (200-hour) endurance test only).
(p) Air temperature after engine, °F. 200-hour endurance test only.
(q) Air temperature intake air entrance, °F. (200-hour endurance test only).
(r) Barometric pressure of engine intake air, in Hg.
(s) Specific fuel consumption, lb/bhp/hr. (gross bhp).
(t) Specific oil consumption, lb/bhp/hr. (gross bhp).
(u) Ignition timing-degrees BTDC. (200-hour endurance test only).
(v) The following shall be recorded at 2-hour intervals when any tests for full load and full throttle are being conducted:

1. Barometric pressure, inches Hg.
2. Temperature, wet and dry bulb, °F.

4.8 Measuring and inspection of testing equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.8.1 Inspection. When inspecting the supplier's inspection equipment, the Government inspector will determine that the supplier has available and utilizes correctly, gaging, measuring, and test equipment of required accuracy, and that the instruments are of proper type and range to make measurements of the required accuracy. The supplier will have available a set of master gages, standards, and appropriate instruments for calibration and these will be maintained by the supplier and made available for review by the Government. Periodically, calibration of gages, standards and inspection equipment will be checked for compliance to gage requirements.

5. Preparation for delivery.

5.1 Preservation, packaging, and packing. Level of preservation, packaging, and packing shall be as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The overhauled engines covered by this specification are intended for use as spares or as replacements in military combat and tactical transport vehicles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title number and date of this specification.
(b) Type of engine required (see 1.2 and 3.1).
(c) Form and number of copies of log sheet to be furnished for overhauled engines (see 3.4.2).
(d) Selections of applicable level of preservation, packaging and packing (see 5.1).
(e) Corrections for cooling-air temperature, if permitted (see 6.3).

6.3 Cooling-air temperatures. Power and cylinder temperatures correction for cooling-air temperature may be omitted. This correction should be determined by test over the applicable temperature range. Such correction, as approved by the procuring activity, should be provided for in the contract (see 6.2).

6.4 Bare engine. The term "bare engine" is defined as an engine without air cleaners, mufflers, cooling fans, with generator unloaded, and less all other power-consuming accessories not considered essential for the operation of engine. As a matter of convenience, the manufacturer may conduct the break-in run using engine cooling fans, and will be allowed to add fan horsepower as approved by the responsible Ordnance Engineering activity to the observed brake horsepower, to arrive at the gross horsepower of the engine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.5 Correction of observed data. Values of torque and horsepower of engines should be corrected for a standard atmospheric conditions by means of the following formulas.

(a) For naturally aspirated engines only

$$CV = OV \times \frac{29.92}{BP-VP} \times \sqrt{\frac{460 + DT}{520}}$$

(b) For supercharges engines only

$$CV = OV \times \frac{29.92}{BP-VP} \times \left(\frac{460 + DT}{520}\right)^{3/4}$$

NOTE: CV = Torque or horsepower corrected for standard atmospheric conditions.

OV = Torque or horsepower actually observed.

BP = Barometric pressure inches of mercury corrected for temperature of carburetor or throttle air entrance.

VP = Water vapor pressure, inches of mercury.

DT = Dry bulb temperature of air intake, °F.

Custodian: 1
Army - AT

Preparing activity:
Army - AT
Project No. 2505 - A502

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DoD-314

OFFICIAL BUSINESS

Commander
US Army Tank-Automotive Command
ATTN: AMSTA-RSES
Warren, MI 48090

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel (either Government or contractor) involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-E-62170(AT)

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4063 WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.